

DEVELOPMENT OF PC-BASED GAME APPLICATION: LION DANCE DASH

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Abstract

Lion Dance Dash is a 2D side-scrolling PC-based game application that brings the vibrant and traditional art of lion dancing to the game realm. This immersive gaming experience combines elements of action, pixel art, and cultural exploration, providing players with an engaging journey through the captivating world of lion dance. In this game, players have to choose the desired dancing lion character to avoid the Nian (enemy) and the obstacles as well as collecting all the coins along the game to complete the level. There will be 3 levels where the amount and type of enemies increases as the level gets higher. Players also can choose to ignore or kill the Nian.

1. Game Title

The project title for Game Development was named Lion Dance Dash. The title was chosen because lion dances are a traditional performance that takes place during Chinese New Year (CNY). Lion dances, also known as múa lân (Vietnam) or barongsai (Indonesia), are performed to ward off evil spirits and bring good luck in the New Year. Therefore, if you hear rhythmic drumbeats and see beautiful lions dancing along the street, it means that Lunar New Year has arrived.

2. Game Logline

Lion Dance Dash is a 2D side-scrolling PC-based game application where players control the movements of the lion dance character to collect money and kill Nian Shou, the mythical monster. Players can choose one out of the three different lion dance characters. Each has different abilities and must collect all the coins and defeat or avoid all the enemies as well as reach the final destination to complete the level.

2.1. Game Lore

The Nian Shou is a mythical creature that is said to emerge from the sea on Chinese New Year to devour people and livestock and terrorise the village. The dancing lion can kill the Nian Shou with their unique abilities and protect the village and the people. When the lion finally wins, people can celebrate Chinese New Year peacefully.

3. Problem Statement

The existing application does not convey the myth behind the lion dance. The existing application may be solely concerned with the physical representation of the lion dance without giving users context about its cultural and historical value. Users, particularly those from non-Asian cultures or unfamiliar with lion dance traditions, may struggle to properly understand the dance without context. The application includes information about the cultural significance of the lion dance once users finish each level that improves the overall user experience and gives educational value.

The existing application can only run when there is an active internet connection. When internet connectivity is inconsistent or unavailable, users may suffer accessibility issues. This constraint may disrupt the user experience, particularly in places with weak network coverage or while travelling where access is inconsistent. Thus, enhancing the application's offline capabilities should be considered. Addressing these concerns can lead to a more user-friendly and versatile application, ensuring that users have a consistent and reliable experience regardless of their internet access state.

4. Objective

- To develop a PC-based game application that promotes traditional Chinese culture, specifically the story behind the lion dance.
- To develop an offline PC-based game application.
- To perform functional testing and user acceptance tests on the developed application to the target user.

5. Gameplay Style

Lion Dance Dash is a single-player 2D desktop side-scrolling game delivered in pixel based. It is an action game of the platformer genre where to move the dancing lion in an environment with obstacles and enemies. The player must use the strength of the dancing lion to attack the Nian Shou of different abilities. The level gets harder as the number and type of enemies increases. The dancing lion needs to collect all the coins, kill, or avoid all the enemies and reach the final destination to complete the level.

6. Intended Audience

"Lion Dance Dash" is an exhilarating game designed for a broad audience, targeting players of all ages. With its ESRB rating [1] of "E" (Everyone), the game ensures a family-friendly and inclusive gaming experience. The intended audience includes children, teenagers, and adults alike, fostering a sense of excitement and joy for players of various backgrounds. Whether you're a casual gamer looking for a delightful pastime or a parent seeking a safe and entertaining game for your children, "Lion Dance Dash" caters to a diverse demographic, offering a visually appealing and culturally rich gaming environment suitable for players of all ages.

7. Concept Art

Concept art in game development is a crucial aspect that involves the creation of visual representations of game concepts and ideas. It is a process that requires creativity and problem-solving, as it involves the construction of new concepts and the creation of the game world [2].

7.1. Character Design

This section provides a detailed discussion of the six characters that are crucial to the game's storyline. Red, Swordie, and Leaper are the protagonists and serve as the heroes, while Flamethrower, Speedster, and Bitey are the antagonists and serve as the villains. Their character's skills will be displayed on the storyboard.

Figure 7.1, Figure 7.2, and Figure 7.3 show the characters design as the heroes while Figure 7.4, Figure 7.5 and Figure 7.6 show the characters design as the villains.



Fig 7.1 Red



Fig 7.2 Swordie



Fig 7.3 Leaper



Fig 7.4 Flamethrower



Fig 7.5 Speedster



Fig 7.6 Bitey

8. Game Flowchart

Game Flow, a model of player enjoyment, has been widely used in game design and evaluation [3]. It has been applied to multiplayer games, with a study finding that elements such as Challenge and Control can vary in this context. The model has also been used in the evaluation of an online strategy game, alongside player experience measures [4]. In the context of education, board games have been proposed as a method for studying flow charts, with the aim of intensifying the learning of algorithmic structures [5]. These studies collectively highlight the versatility and potential applications of the game flow model in game design, evaluation, and education.

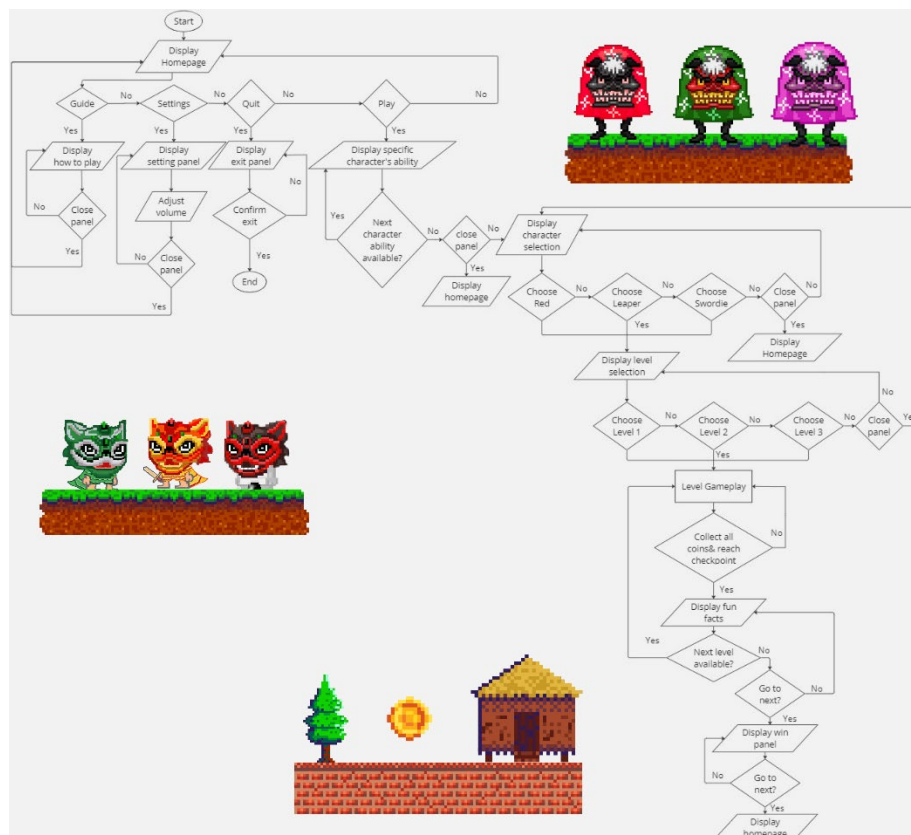


Fig. 8.1 Game flowchart

9. Navigation Structure

This section pertains to the navigation structure [6] of the proposed application, 'Lion Dance Dash'. It details how the different interfaces of the application are organized and connected to one another. The navigation structure is crucial for developers to check the overall content structure of the application by providing an easily understandable roadmap of all interfaces. Figure 8.1 illustrates the navigation structure for the current project, detailing the interface progression from the homepage to each available option.

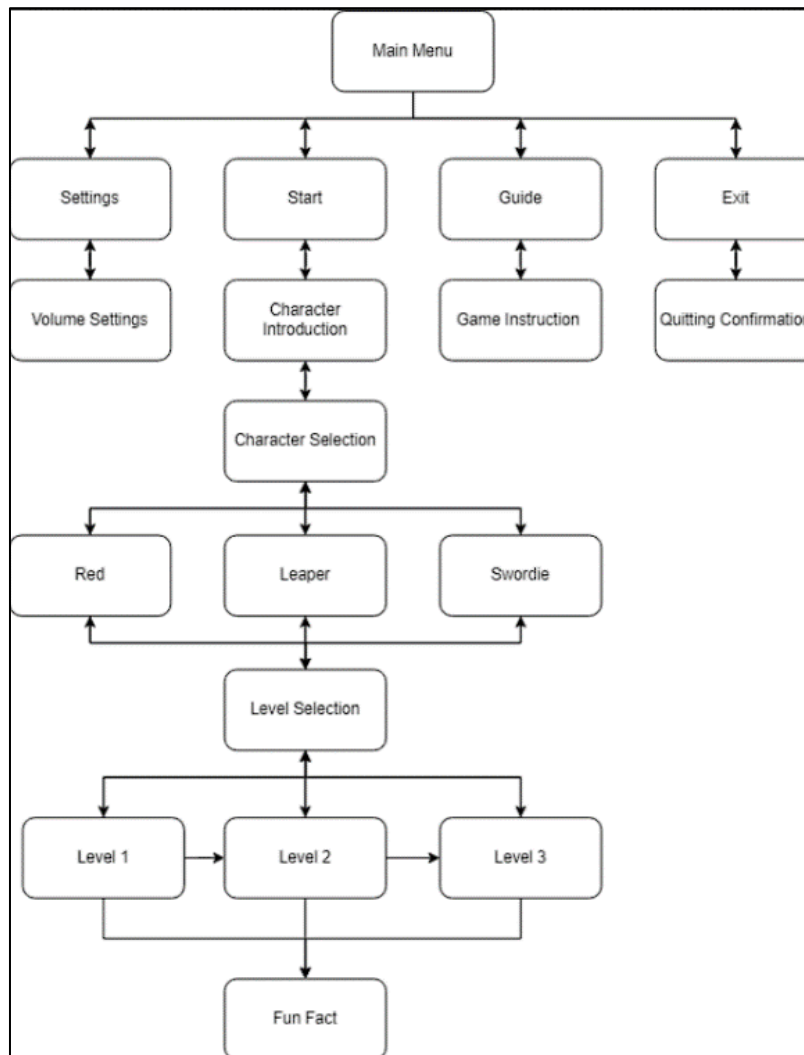


Fig. 9.1 Navigation Structure

10. Storyboard

Storyboarding is where you take your groundbreaking gameplay ideas and reproduce them in a visual storytelling format [7]. It's super useful when your game design has lots of steps or focuses on a particular visual presentation. Design of interfaces such as the homepage interface, main menu interface, storyline interfaces, gameplay interfaces, and setting interface are included in the section below

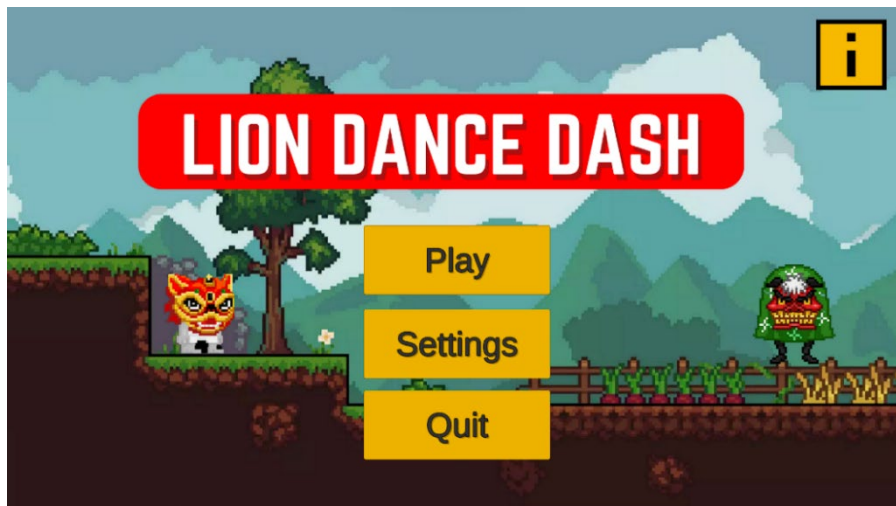


Fig. 10.1 Homepage Interface

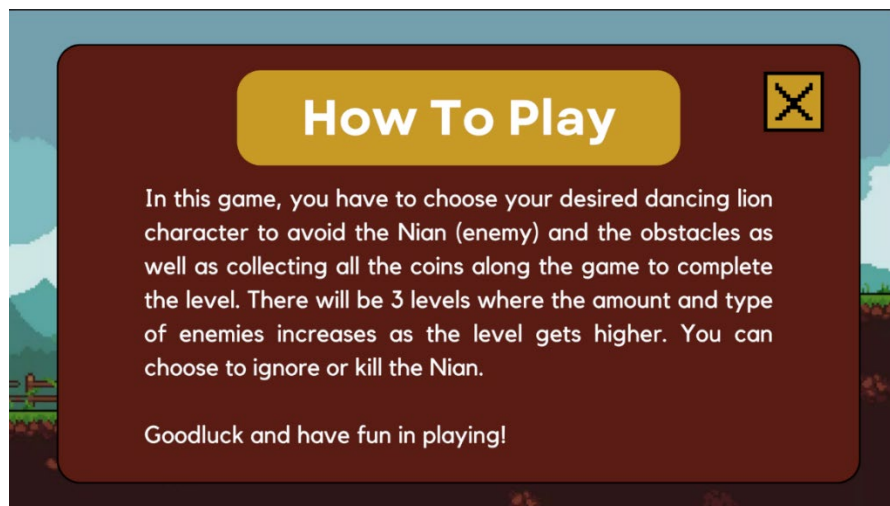


Fig. 10.2 User Manual

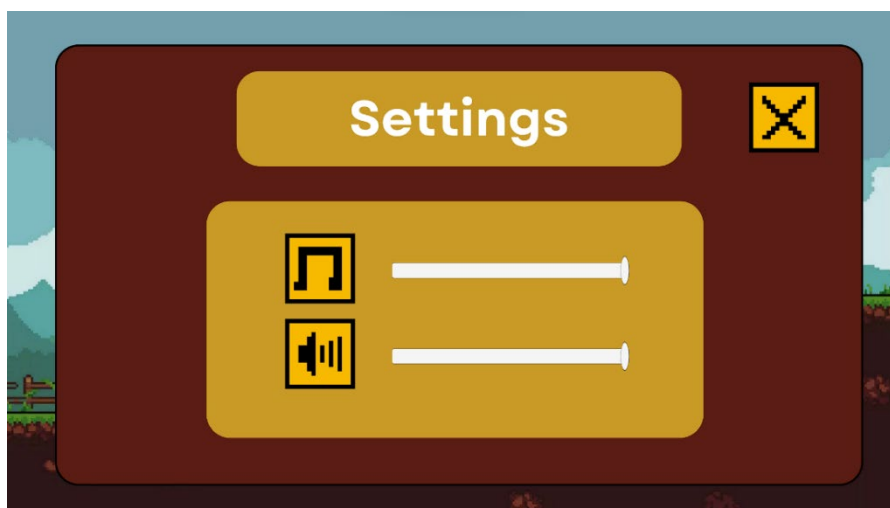


Fig. 10.3 Settings Interface

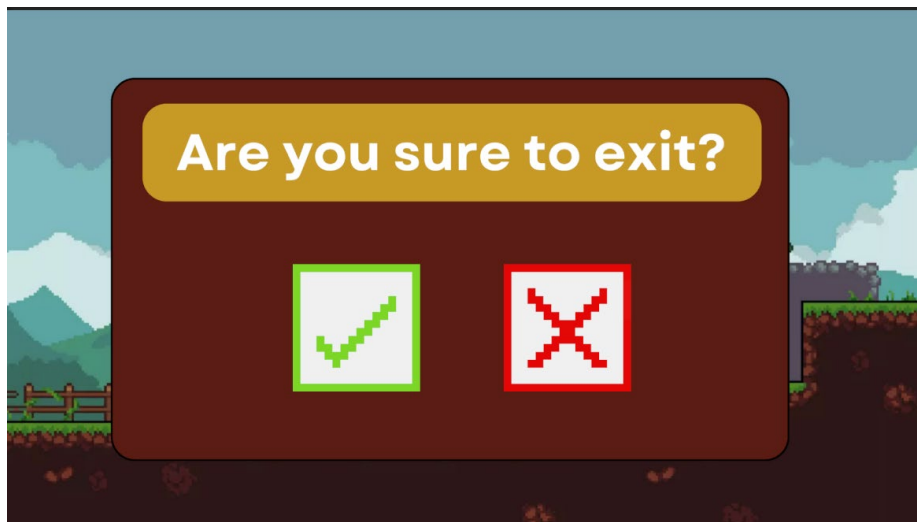


Fig. 10.4 *Exit Confirmation Interface*



Fig. 10.5 *Character "Red" Description Interface*

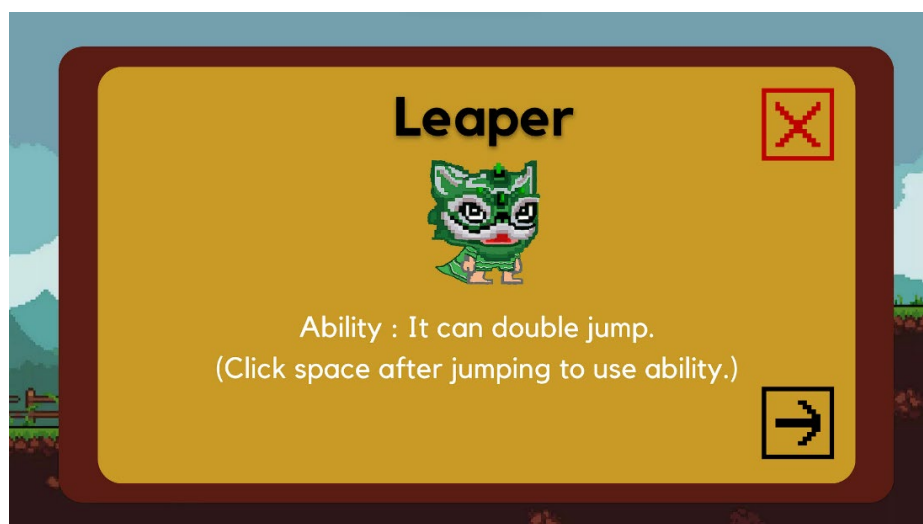


Fig. 10.6 *Character "Leaper" Description Interface*

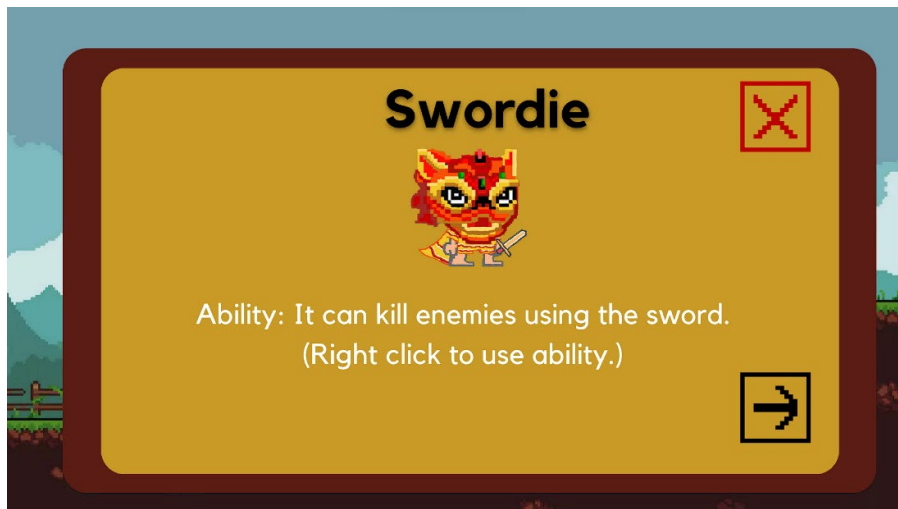


Fig. 10.7 Character "Swordie" Description Interface

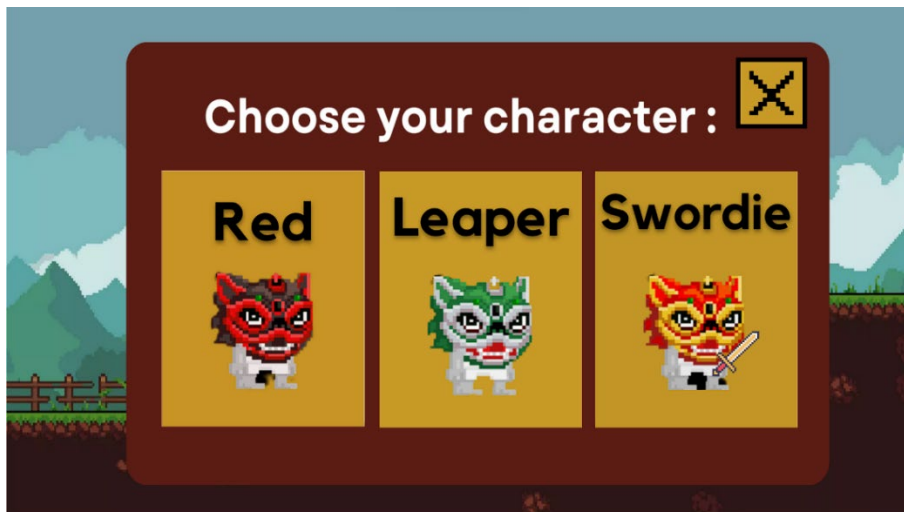


Fig. 10.8 Character Selection Interface

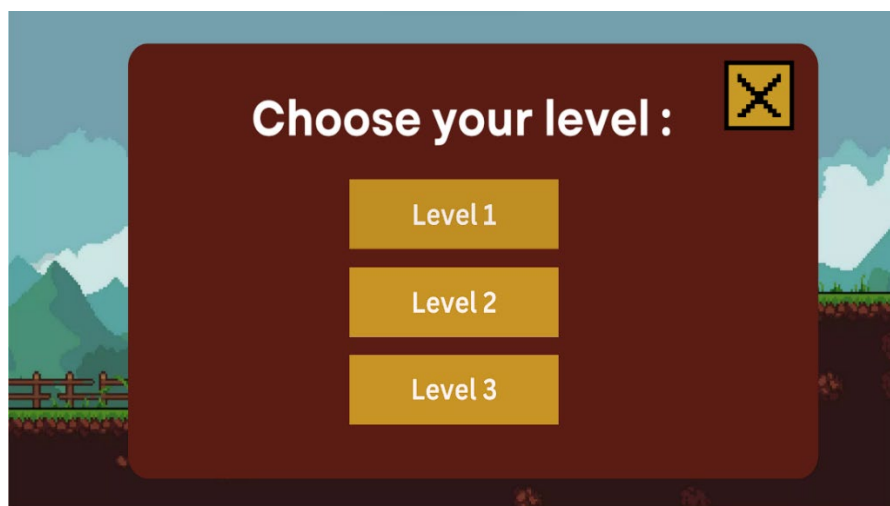


Fig. 10.9 Level Selection Interface



Fig. 10.10 Fun Fact Level 1 Interface

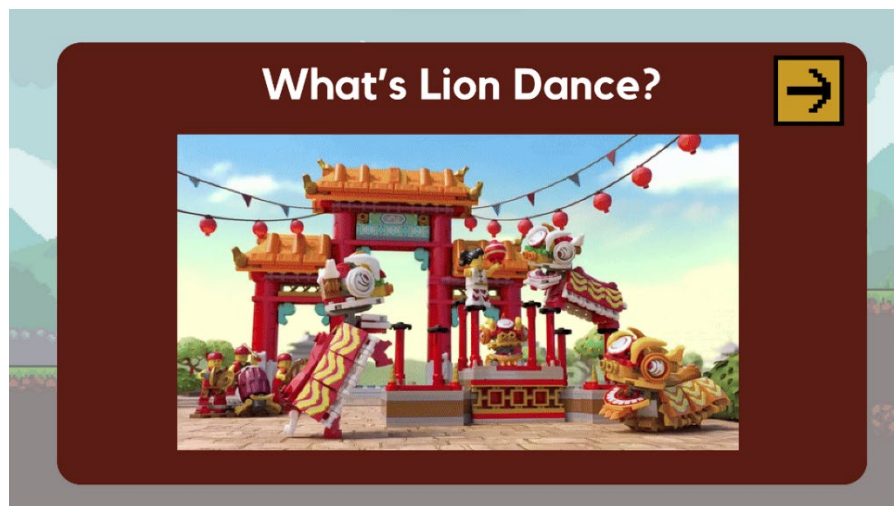


Fig. 10.11 Fun Fact Level 2 Interface



Fig. 10.12 Fun Fact Level 3 Interface

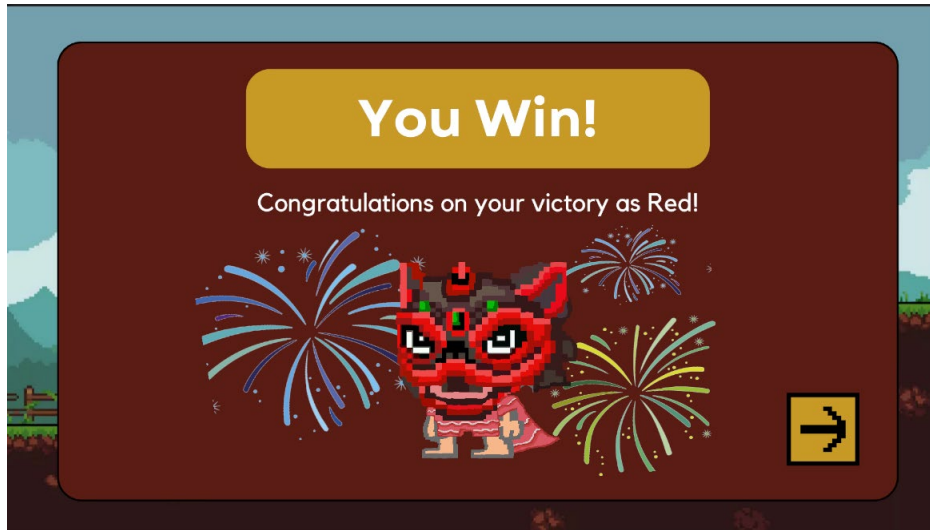


Fig. 10.13 *Sample of Game Completion Interface*

11. Level Selection

This section will discuss the levels of the game, there are a total of three levels in the game, each with its own difficulty level. As players progress through the game, the variety of Nian Shou will increase and the difficulty will also increase.



Fig. 11.1 *Level 1*



Fig. 11.2 *Level 2*



Fig. 11.3 *Level 3*

12. Game Samples

Our project team update and innovate this existing game to promote the culture of lion dance and let more people know about it.

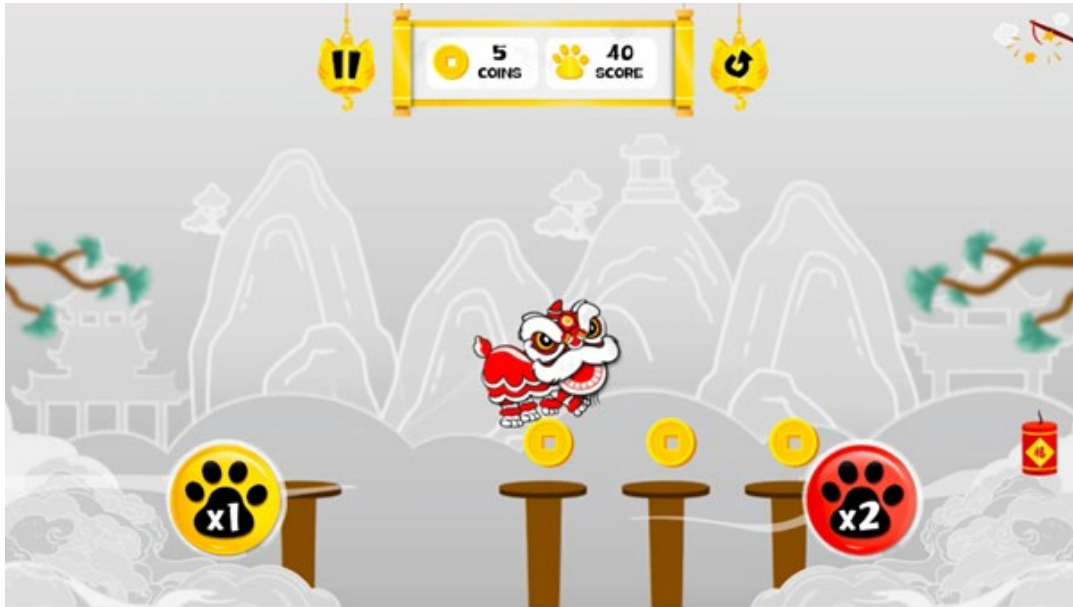


Fig 12.1 Game sample of the game *Lunar Lion Dance 2023* by Mason Games available in Google Play Store.

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Conflict of Interest

Authors declare that there is no conflict of interests regarding the publication of the paper.

Author Contribution

This journal requires that all authors take public responsibility for the content of the work submitted for review. The contributions of all authors must be described in the following manner:

*The authors confirm contribution to the paper as follows: **study conception and design:** Er Jia Wei, Lee Bing Yang; **data collection:** Lee Ling Yi, Steven Ngu Chung Kiet **analysis and interpretation of results:** Lee Ling Yi, Loo Qiao Li; **draft manuscript preparation:** Er Jia Wei, Lee Bing Yang, Lee Ling Yi, Steven Ngu Chung Kiet, Loo Qiao Li. All authors reviewed the results and approved the final version of the manuscript.*

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